CS315 · ALGORITHMS ANALYSIS & DESIGN

Midterm 461 Practice

Midterm 461 · 25 marks

Muharib

01 About this midterm

The midterm of term 461: five questions out of 25. The questions are copied from photos of a student's paper, and his answers are covered in most of the photos.

Question	1	2	3	4	5	Total
Marks	3	6	5	6	5	**25**

About the solutions

I have no official answer key. In Q2 the student's choices are visible in the photos, and I checked them and found them all correct. The other questions are solved by me, and the tables are computed by code.

02 Questions

Q1 TRUE / FALSE · 3 PTS

Choose True or False by putting a check mark (✓):

1) An algorithm is a disorganized, unordered set of steps to solve a problem in an unspecified amount of time.

2) The running time of an algorithm typically grows with the input size.

3) Insertion Sort is an algorithm that correctly sorts an arbitrary n-element of an array in time about n.

4) The main operations of a dictionary ADT are searching, inserting, and deleting items.

5) Fractional knapsack problem can be solved in time O(n)

6) In any problem there can be many optimal solutions but there is only one feasible solution

Q2 MCQ · 6 PTS

Choose the correct answer by putting a check mark beside the correct answer:

1) When we apply the Merge sort algorithm to the sentence S=(F,I,N,A,L,E,X,A,M) to sort it into increasing order, the first two sub-problems are:

- ☐ (F,I,N) and (A,L,E,X,A,M)
- ☐ (F,A,L,E,I) and (N,X,A,M)
- ☐ (F,I,N,A,L) and (E,X,A,M)
- ☐ (F,I,N,A) and (L,E,X,A,M)

2) In the Dictionary ADT, the log file is most effective when:

- ☐ Only searches are frequently performed
- ☐ Searches and insertions are equally common
- ☐ Both insertions and removals are frequent
- ☐ Insertions are frequent and searches are rare

3) What is the first step in developing a dynamic-programming algorithm?

- ☐ Compute the value of an optimal solution
- ☐ Characterize the structure of an optimal solution
- ☐ Construct an optimal solution from computed information
- ☐ Recursively define the value of an optimal solution

Q2 MCQ · CONTINUED

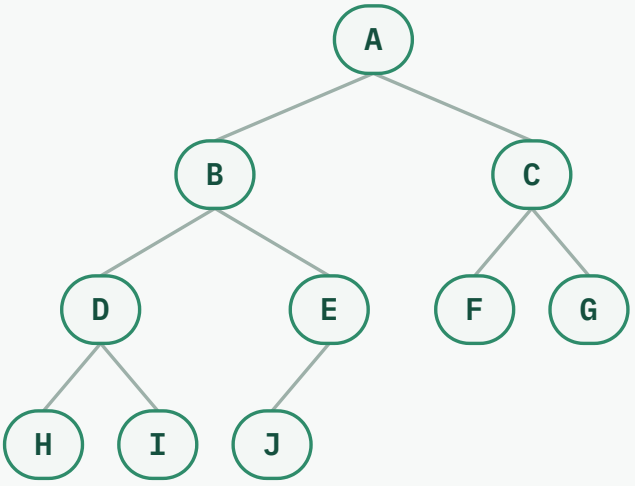
4) Given two strings: s1="AGGTAB" and s2="GXTXAYB", the Longest Common Subsequence (LCS) using dynamic programming is:

☐ 3 ☐ 4 ☐ 5 ☐ 6

5) What is the running time of the Huffman encoding algorithm?

☐ O(log n) ☐ O(n) ☐ O(n log n) ☐ $O(n^2)$

6) From the following given tree, what is the code word for the character 'J'? (a left edge is 0 and a right edge is 1)



☐ 101 ☐ 110 ☐ 010 ☐ 001

Q3 DOUBLE HASHING · 5 PTS

Consider a hash table with size 10, show the hash table after all items have been inserted.

- The primary hash function is h(k) = k mod 10.
- The collision resolution strategy is double hashing.
- The secondary hash function is d(k) = k mod 9.
- Key to insert {19, 16, 90, 18, 60, 89, 70}

K	H(k)	D(k)	Probes
19			
16			
90			
18			
60			
89			
70			

0	1	2	3	4	5	6	7	8	9

Q4 MATRIX CHAIN · 6 PTS

Apply the Dynamic programming algorithm and give the optimal solution to the matrix chain multiplication ABC where the order for matrix,

A = 5 × 10, B = 10 × 3, C = 3 × 12.

Note: Provide the best order of matrix multiplication and the number of operations.

Minimum operations = _____ Best order of multiplication: _____

	A	B	C
A			
B			
C			

Q5 FRACTIONAL KNAPSACK · 5 PTS

Given items as {value, weight} pairs {{40,20},{30,10},{20,5}}. The capacity of knapsack=20. Find the maximum value output assuming items to be divisible.

The maximum value = _____